ENVIRONMENTAL SOUND CLASSIFICATION / REPORT 02

CNN and CRNN baselines

Complete historical study results

The earlier frozen baseline comparison: repeated seeds, reserved-fold results, paired errors, order probes, synthetic stress and fixed-recipe fold rotation. Historical test results remain historical; this PDF export performs no new evaluation.

21	3	8
Full neural runs	Repeated training seeds	Embedded figures

Prepared 29 October 2026. Main recorded experiments: 28 October 2026.

All figures are embedded at their original resolution in the full-page figure atlas. Click any figure reference to jump to its page. Full tables and scientific limitations are retained. Companion-project evidence paths are indexed at the end.

This report documents internal agent-produced execution and verification. It does not certify external human replication, learner mastery, state of the art or production readiness.

Scope note for this 29 October export

The body preserves the 28 October completed study and its then-current 153-test verification scope. The later matrix study raised project checks to 274 and re-executed additional development tasks; those belong to Report 01. The two latency tables also use different hardware/workloads. Keep their scopes separate.

Completed study, 28 October 2026. Agent-produced research and implementation, with separate internal verification. This is a from-scratch comparison on one public dataset; it makes no new-architecture, state-of-the-art, institutional-affiliation or learner-mastery claim.

Findings and delivery decision

The small CNN is the frozen default. CNN+GRU improved mean development accuracy by **1.26 percentage points**, below the predeclared 3-point practical preference. On the reserved fold 5, their three-seed means were **63.08% and 63.92%**. The primary seed 101 checkpoints scored **61.50% and 62.00%**. These results establish a working classifier and modest recurrent gains within this setup; they do not establish recurrent superiority.

The model is efficient under the measured workload, but **noise robustness was not achieved**: the default CNN fell to 42.50% accuracy at 20 dB Gaussian-noise SNR and 6.25% at 0 dB. Its primary fold 5 airplane recall was 0/8. The deliverables are research prototypes with these limitations made explicit.

A fixed-recipe, one-seed rotation across five outer folds gave 66.50% CNN and 68.30% CNN+GRU mean accuracy. Folds 1-4 had already informed development, so this is descriptive characterization rather than five untouched confirmations or a fully nested estimate of the adaptive development process. No model selection was changed after test access.

Question, hypotheses and scope

The question was whether a bidirectional GRU adds enough value to justify its extra parameters and computation when the frontend, encoder, actual training clips, maximum exposure and checkpoint rule are fixed. The competing explanation is that local time-frequency texture and nonlinear nonrecurrent heads already provide enough information. Extra capacity, optimization and selection variation can also explain an apparent gain.

The study compares a small pooled CNN, a parameter-comparable pooled MLP, a shared tokenwise MLP and a CNN+GRU. The tokenwise model applies its nonlinearity before averaging; it is a stronger nonrecurrent control than only enlarging an already pooled head. Equal parameter counts do not match FLOPs, dropout granularity or optimization. An RNN-only model was considered in the initial source audit but was not part of the final controlled study; the recurrence question here concerns aggregation on a shared convolutional encoder.

These are established families. Qiao et al. 2019 provides a directly relevant historical CNN/CRNN comparison; Çakır et al. 2017 motivates retaining frequency identity before recurrence, on a different detection task. Salamon and Bello 2017 studies augmentation on UrbanSound8K. Their reported scores are not interchangeable with this protocol. The full-text audit [P01] records reading depth and unresolved source limitations; it is not an exhaustive 2026 novelty search or a reproduction of those papers.

Data and evaluation boundary

ESC-50 is pinned to commit `33c8ce9eb2cf0b1c2f8bcf322eb349b6be34dbb6`: 2,000 five-second recordings, 50 categories and five official folds of 400 clips. No external pretrained weights or data were used. The official metadata and license are retained; the dataset is distributed overall under CC BY-NC 3.0. See the official repository.

Four source IDs cross official folds. IDs 131943/134049 cross folds 2/3; 209698/234879 cross 4/5. These establish source dependence risk, not proof of identical or overlapping waveform content. The strict protocol retains the entire test fold, removes test sources from validation and removes held-out sources from training. Development therefore used 1,200 training clips, 398 validation clips in fold 4, and 400 reserved fold 5 clips. Normalization was fitted on the actual training subset. Format and exact-byte-duplicate audits found no exact cross-fold WAV duplicate; perceptual near-duplicate auditing remains incomplete.

Deterministic features were cached for all clips before testing, but held-out statistics were not fitted and test predictions did not inform selection. The selected model, seed, checkpoint identities, probe settings, fold roles and resource limits were saved in confirmation_v1.json [P02]. Independent review found the first recorded holdout access 98.55 seconds after this freeze. The fixed confirmation seed was101, rather than the best observed test seed.

Primary metric is clip top-1 accuracy; macro-F1 averages all 50 class F1 values with undefined values set to zero. Every checkpoint is the earliest minimum validation cross-entropy. No later maximum-accuracy checkpoint was substituted. Each changed fold 5 decision changes accuracy by 0.25 pp. See the source audit [P03] and data audit [P04] and waveform audit [P05].

Method and training

The frontend downmixes to mono, resamples to 16 kHz, uses a 512-sample Hann STFT with 256-sample hop, forms 64 HTK mel-power bands and takes the natural log with floor 1e-10. Train-only per-band normalization produces `[B,1,64,313]` features. Three conv/BatchNorm/ReLU/pooling stages produce `[B,64,8,39]`.

Frequency-preserving heads retain the channel×frequency dimensions. The CNN averages 39 time tokens into 512 features; the hybrid sends 39×512 tokens to a bidirectional GRU with hidden size 64 per direction, averages its 128-dimensional outputs and projects to 50 logits. The method document [P06] gives equations, tensor shapes, receptive-field assumptions, GRU parameter arithmetic, augmentation and SNR checks.

All registered repeated models used AdamW (learning rate 0.001, weight decay0.0001), batch size 32, gradient clipping 5,60 epochs and cosine decay, with no augmentation. Every development run saw 72,000 training examples and 2,280 optimizer updates. Seeds 101/202/303 vary initialization and data order; they are three runs of the same task, not independent datasets. The statistical anchor concatenates 64 temporal means and 64 population standard deviations, standardizes using training data, and fits multinomial logistic regression. A declared L2 grid 0.001/0.01/0.1 selected 0.001 by validation cross-entropy; optimization was extended to convergence before final use.

An initial six-model screen at seed 17 motivated the frequency-preserving family. Its saved notebook and complete curves survive, but its model weights and row predictions were not recovered before the temporary runtime ended. Its 75.38% validation result is exploratory history, not a recoverable final checkpoint. The final 13 development runs and eight additional fold-training runs all have saved portable selected weights and complete records. The augmentation pilot reached 73.12% validation accuracy versus 75.13% for its unaugmented seed 101 counterpart, failing the predeclared 2 pp improvement gate; further augmentation repeats were therefore not run.

Repeated results and selection

Values below are percentages. Neural ± values are **sample SD over three training seeds**, not confidence intervals. Each seed uses the same398 validation and 400 test clips. The deterministic statistical baseline is a single selected fitted model.

Model	Parameters	Validation accuracy	Fold5 accuracy	Fold5 macro-F1
CNN	49,058	72.61 ± 0.25	63.08 ± 1.63	61.77

Model	Parameters	Validation accuracy	Fold5 accuracy	Fold5 macro-F1
CNN + GRU	251,810	73.87 ± 1.26	63.92 ± 1.77	62.04
Matched pooled MLP	252,036	72.28 ± 0.63	61.50 ± 3.36	59.38
Tokenwise MLP	252,036	69.68 ± 3.90	59.33 ± 4.02	57.35
Temporal statistics + logistic regression	6,450	42.21	37.25	35.04

Figure 1. Model comparison - see the full-page figure atlas.

The larger nonrecurrent heads did not improve on the small CNN. The recurrent model's validation gain was 1.26 pp and its held-out mean gain 0.83 pp. The registered preference retained the best eligible nonrecurrent model unless recurrence improved mean validation accuracy by at least 3 pp without lowering mean macro-F1. All models met the prototype latency/size limits, so the default remained `cnn_freq`. This is an operational decision under a stated utility rule, not a universal conclusion that smaller gains have no value.

The validation-to-test drop is material: the selected CNN family's mean decreases from 72.61% to 63.08%. Learning curves show high training accuracy and substantially lower validation accuracy. Selection optimism and fold-specific recording differences are plausible contributors; these observations do not isolate their causes. We did not respond by tuning on fold 5.

Figure 2. Learning curves - see the full-page figure atlas.

All individual neural runs are retained:

Model	Seed	Validation accuracy	Fold5 accuracy	Fold5 macro-F1	Selected epoch
CNN	101	72.61	61.50	59.92	47
CNN	202	72.36	64.75	63.71	49
CNN	303	72.86	63.00	61.67	46
CNN + GRU	101	75.13	62.00	59.50	58
CNN + GRU	202	73.87	65.50	63.82	58
CNN + GRU	303	72.61	64.25	62.80	30
Matched pooled MLP	101	71.61	57.75	55.54	32
Matched pooled MLP	202	72.86	62.50	60.46	52
Matched pooled MLP	303	72.36	64.25	62.15	48
Tokenwise MLP	101	65.33	54.75	51.40	13
Tokenwise MLP	202	72.86	62.25	60.99	30
Tokenwise MLP	303	70.85	61.00	59.65	20

Paired errors and uncertainty

At seed 101 the CNN and CRNN were both correct on 213 clips, CNN-only correct on 33, CRNN-only correct on 35, and both wrong on 119. Thus the recurrent primary checkpoint adds only two net correct decisions. A paired source-group bootstrap (310 groups, 2,000 resamples, seed 42, clip-weighted accuracy) gives a 95% percentile interval of **-3.90 to +4.89 pp for CRNN minus CNN**. This is fixed-model sample-composition uncertainty. It excludes retraining, fold and model-selection uncertainty; the three-seed SDs above describe a different source of variation. Neither the point difference nor this interval establishes superiority or equivalence.

The selected CNN correctly classified **0/8 airplane clips**. The primary CRNN had 0/8 recall for airplane and fireworks. Other named confusions are shown below. Eight examples per class support concrete failure inspection but give imprecise class-level estimates. No corrective tuning followed these observations. All 400 test clips were above the silence threshold, so abstention did not alter test coverage.

The largest primary-checkpoint class-count differences are listed below (ties sorted by category name). These are exploratory slices with eight examples per class, not independent evidence of a temporal mechanism.

Category	CNN correct	CRNN correct	CRNN minus CNN
footsteps	2/8	5/8	+3
siren	2/8	5/8	+3
clapping	2/8	4/8	+2
coughing	8/8	5/8	-3
train	7/8	4/8	-3
door wood creaks	3/8	1/8	-2

Figure 3. Named confusions - see the full-page figure atlas.

Temporal-order probe

Five deterministic permutations were applied after the encoder. All nonrecurrent controls retained identical predictions, as expected. CRNN predictions changed on 34-45 of 400 test clips, while accuracy ranged 61.00-62.25% around clean 62.00%; the mean change was -0.60 pp. This establishes order dependence of some predictions under an artificial intervention, but provides weak evidence that the ordering improved classification accuracy. Shuffled tokens are out of distribution; the probe cannot by itself establish a causal mechanism for a natural-audio improvement.

Figure 4. Order probe - see the full-page figure atlas.

Synthetic noise, gain and input behavior

Noise was Gaussian and scaled to each processed waveform's RMS at the registered 20/10/0 dB SNR, with seed 2026 and no peak clipping. Gain changes were -6/+6 dB. These are synthetic-corruption measurements, not proof of real-world microphone or background robustness.

Condition	Primary CNN accuracy	Primary CRNN accuracy
Clean	61.50	62.00
20 dB SNR	42.50	45.25

Condition	Primary CNN accuracy	Primary CRNN accuracy
10 dB SNR	24.50	25.00
0 dB SNR	6.25	7.00
Gain -6 dB	59.50	60.50
Gain +6 dB	57.75	61.50

Figure 5. Synthetic stress - see the full-page figure atlas.

Both models degrade severely under additive noise. The clean-validation augmentation pilot failing its gate does not show that augmentation cannot improve noise robustness; its held-out noise performance was not used for selection. The input interface rejects invalid/nonfinite or overlong recordings, pads shorter ones and abstains when waveform RMS < 1e-5. Softmax scores are uncalibrated, and arbitrary unknown sounds are not detected. Short inputs are accepted by preprocessing but accuracy was evaluated on five-second ESC-50 clips.

Five-fold fixed-recipe characterization

After development and confirmation were frozen, the two original frequency-preserving architectures were trained with the same 60-epoch recipe at seed 101. Each outer fold uses the preceding fold cyclically for validation and the remaining three for training, with source repair. The existing fold 5 seed 101 models/results were reused. All test folds retain 400 clips.

Outer test fold	Validation fold	Train / validation clips	CNN accuracy	CNN+GRU accuracy
1	5	1198 / 400	64.75	71.00
2	1	1198 / 400	65.00	67.25
3	2	1200 / 398	68.00	69.25
4	3	1195 / 400	73.25	72.00
5	4	1200 / 398	61.50	62.00
Mean across folds	-	-	**66.50**	**68.30**
Sample SD across folds (pp)	-	-	4.42	3.96

Figure 6. Five-fold characterization - see the full-page figure atlas.

The mean difference is 1.80 pp, with substantial fold variation. These are source-strict three-training-fold results, not directly comparable with published protocols that train on four folds, use patches, external pretraining or different selection procedures. Folds 1-4 already participated in adaptive development. Five folds share training data and are not five independent datasets. This rotation is descriptive fixed-recipe context; it does not repair the history into a fully nested evaluation or create five untouched confirmations.

Efficiency and resource accounting

CPU measurement used an Apple M4, two threads, batch size 1, 10 warmup and 100 measured repetitions of the same validation WAV. File cache and model were warm. WAV-to-logits includes decoding, resampling, log-mel and normalization; it excludes model loading, CLI startup, softmax/JSON service handling and concurrency. The declared prototype limits were p95 ≤ 100 ms and a compact checkpoint ≤ 5 MiB.

Model	Compact checkpoint bytes	Model-only p95 (ms)	WAV-to-logits p95 (ms)
CNN	286,113	1.33	3.64
CNN + GRU	1,098,142	1.91	4.36
Matched pooled MLP	1,098,587	1.40	3.39
Tokenwise MLP	1,098,587	1.19	3.40

All models meet those narrow limits. Parameter count is not a latency ratio. Fresh standalone CLI checks took about 1.0-1.3 seconds including process startup and model loading, so those should not be confused with the warmed figures. No CUDA-inference or production-service latency claim is made.

The 21 final 60-epoch training runs sum to 1,553.35 seconds of recorded per-run wall time, including their logged epoch evaluation/checkpoint work. This excludes preparation, earlier pilots, failed orchestration attempts, idle allocation, probes, verification and notebook execution. Historical CPU pilots/40-epoch runs and the first six-model screen remain recorded separately; the first screen's six full runs sum to about 459.53 seconds. These totals are not a complete account of all project time or GPU-active compute. No compute-unit purchase, paid API purchase or GPU-plan upgrade was made. Colab used a Tesla T4, PyTorch 2.11.0+cu130, NumPy 2.1.3, SciPy 1.16.3 and SoundFile 0.14.0; the complete environment is saved with the study.

Verification and implementation quality

The implementation now lives in a small `src/esc50_sound` package with explicit audio, data, model, training, evaluation, inference, baseline and record responsibilities. There is one supported CLI and no experiment-framework dependency. Organizational references were the explicit model/train/evaluate structure in PyTorch's MNIST example and data/pipeline separation in torchaudio's wav2letter example. Root launchers preserve old commands; frozen historical code is archived. Independent internal checks found package/frozen-source logits bit-for-bit equal for all four repeated architectures on the same input. Both primary models reproduced every saved validation prediction from 398 original WAVs, and a separate reviewer reproduced all 400 primary test predictions for each model through the packaged frontend/model.

The current suite passes 153 tests. Saved prediction/selection/checkpoint audits pass for all 13 development runs and 20 neural held-out evaluations. The statistical baseline's 400 labels/predictions were independently recalculated. A separate reviewer checked the freeze chronology, primary paired errors, actual WAV inference and absence of test abstention. This is internal independent execution, not external human replication or evidence of learner mastery. Cached-feature bytes depend on the numerical runtime; the frozen environment supports replay, while the raw-WAV delivery checks establish portability across the tested Colab/local versions. Exact resume from the compact deployment checkpoints is not implemented.

The final self-contained Colab notebook was executed end to end on a T4 with a two-epoch integration override, including both training loops, the statistical baseline, real WAV prediction and ZIP export. All 20 code cells completed without errors; their code matches the default 60-epoch template. Those two-epoch demonstration scores are excluded from the research tables. The two primary portable bundles also ran from an unrelated working directory using their own source and empty project PYTHONPATH.

Deliverables and reproduction

• Default CNN model card [P07] and recurrent comparison card [P08].

• Portable model directories [P09], each with weights, source, class ordering, frontend, requirements, model card and file manifest.

• Final Colab notebook, local template [P10] and executed integration record [P11].

• Final study files [P12], development audit [P13], final evaluation audit [P14], paired comparison [P15], and independent verification [P16].

• Figure source [P17] and numeric figure inputs [P18].

From the project directory:

```
python -m pip install -e '.[dev,baseline,report]'
python -m esc50_sound prepare
python -m esc50_sound train --model cnn_freq --seed 101 --epochs 60 --scheduler cosine --device \
  cuda --output runs/new_cnn
python -m esc50_sound predict --checkpoint runs/new_cnn/best.pt --audio YOUR_SHORT_CLIP.wav
python -m pytest -q
python tools/make_figures.py --svg --include-cv
```

Use a new output directory for each run and the saved configuration/environment for exact experimental scope. The notebook requires no opaque source encoding or Drive mount. The full study scripts and frozen source preserve the actual executed recipe; the modern package keeps the model mathematics equivalent while making the implementation easier to navigate. The completion matrix [P19] links requirements to proof.

Limits and research conclusion

This study answers the declared recurrence question within a small from-scratch, single-dataset regime. It supports retaining the simpler CNN under the registered preference and reporting the recurrent model's modest gains alongside its cost. It does not establish universal model ranking, novelty, calibrated probabilities, unknown-sound recognition, short-clip accuracy, robustness on real backgrounds, or production readiness. Near-duplicate auditing is incomplete, three seeds give limited training-variation evidence, and the broader fold estimate has a disclosed development history.

The useful outcome is a reproducible classifier comparison with a negative robustness result, a clear default choice and a portable implementation. Improving it responsibly would require new development/confirmation boundaries rather than reusing these test outcomes for another tuning cycle.

Appendix A. Full primary-checkpoint confusion matrices

Both matrices use the historical primary seed101 checkpoints on the 400 fold-5 clips. Rows are true classes, columns are predicted classes; cells show within-true-class normalized proportions. They supplement the named-confusion figure and were not recomputed for this PDF.

Figure 7. CNN primary fold-5 confusion matrix - see the full-page figure atlas.

Figure 8. CRNN (CNN + BiGRU) primary fold-5 confusion matrix - see the full-page figure atlas.

Appendix B. Class-ID key for the confusion matrices

Class IDs are shared by both models and follow the pinned official ESC-50 metadata. Underscores in category identifiers are shown as spaces for readability.

ID	Category	ID	Category
0	dog	25	footsteps
1	rooster	26	laughing
2	pig	27	brushing teeth
3	cow	28	snoring
4	frog	29	drinking sipping
5	cat	30	door wood knock
6	hen	31	mouse click
7	insects	32	keyboard typing
8	sheep	33	door wood creaks
9	crow	34	can opening
10	rain	35	washing machine
11	sea waves	36	vacuum cleaner
12	crackling fire	37	clock alarm
13	crickets	38	clock tick
14	chirping birds	39	glass breaking
15	water drops	40	helicopter
16	wind	41	chainsaw
17	pouring water	42	siren
18	toilet flush	43	car horn
19	thunderstorm	44	engine
20	crying baby	45	train
21	sneezing	46	church bells
22	clapping	47	airplane
23	breathing	48	fireworks
24	coughing	49	hand saw

Project evidence index

The paths below refer to the accompanying ESC-50 project. They remain useful provenance references when the PDF is shared, but the project files are not embedded in this PDF. Public web references in the report are clickable.

[P01] `docs/literature_deep_read.md`

[P02] `configs/confirmation_v1.json`

[P03] `archive/research_start_and_source_audit.md`

[P04] `results/data_audit.json`

[P05] `results/waveform_audit.json`

[P06] `docs/method.md`

[P07] `docs/model_cards/cnn_freq.md`

[P08] `docs/model_cards/crnn_freq.md`

[P09] `deliverables/README.md`

[P10] `notebooks/ESC50_Research_and_Inference.ipynb`

[P11] `notebooks/ESC50_Research_and_Inference_output.ipynb`

[P12] `results/final_study`

[P13] `results/development_audit/report.md`

[P14] `results/final_evaluation_audit/report.md`

[P15] `results/holdout_primary_comparison/comparison.json`

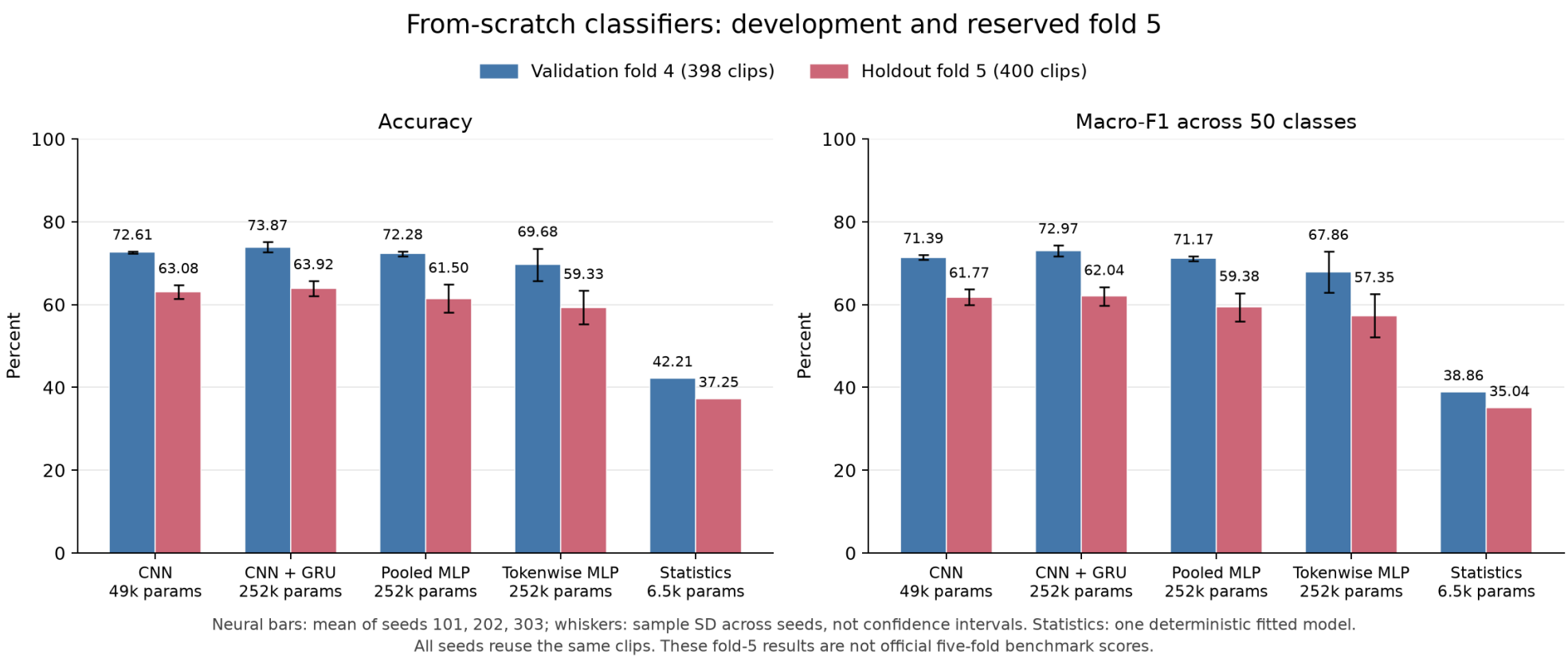
[P16] `results/independent_verification.json`

[P17] `tools/make_figures.py`

[P18] `results/figures/figure_data.json`

[P19] `docs/completion_plan.md`

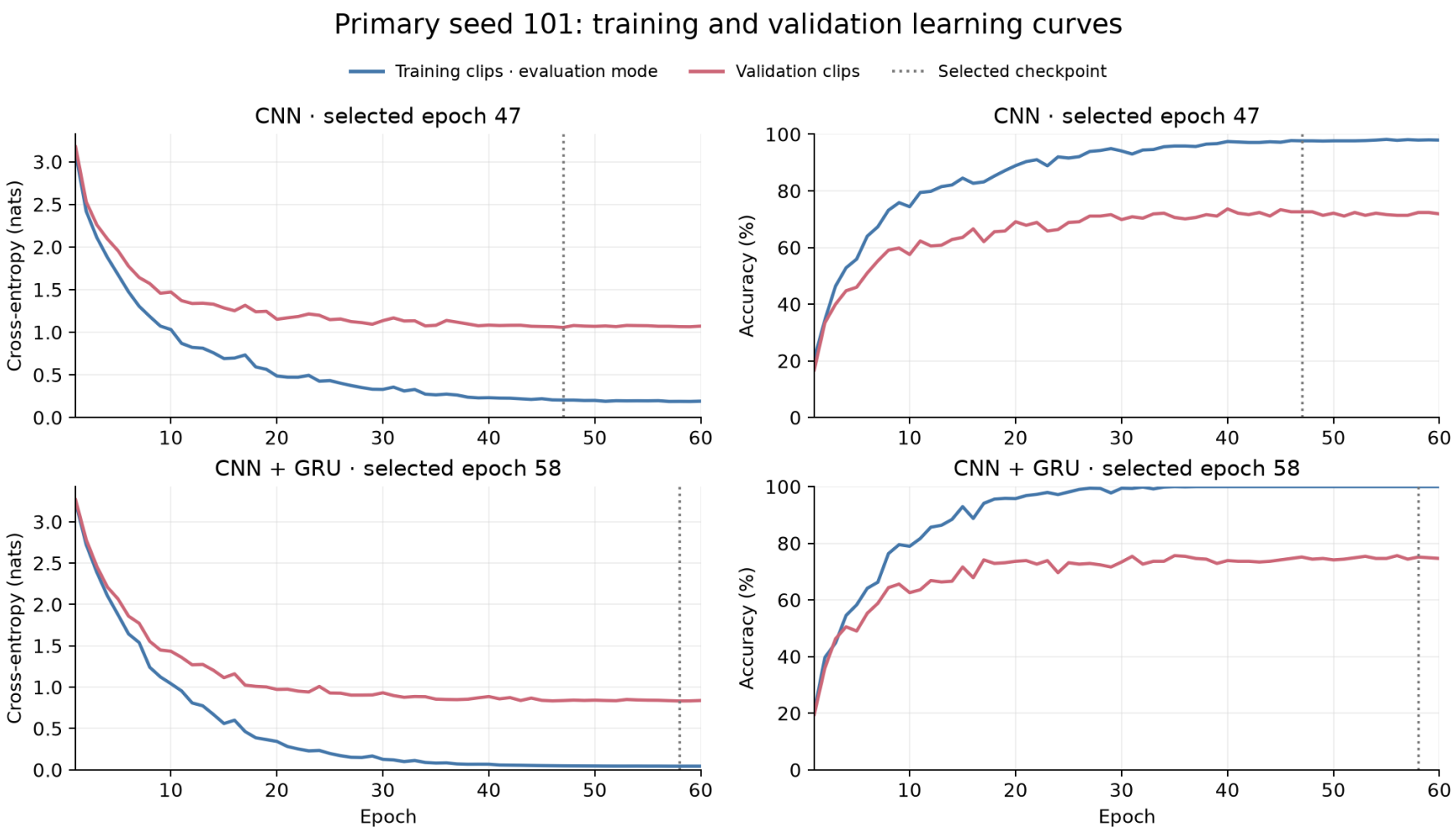
Figure 1. Model comparison



Model comparison

Source: `results/figures/model_comparison.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 2. Learning curves

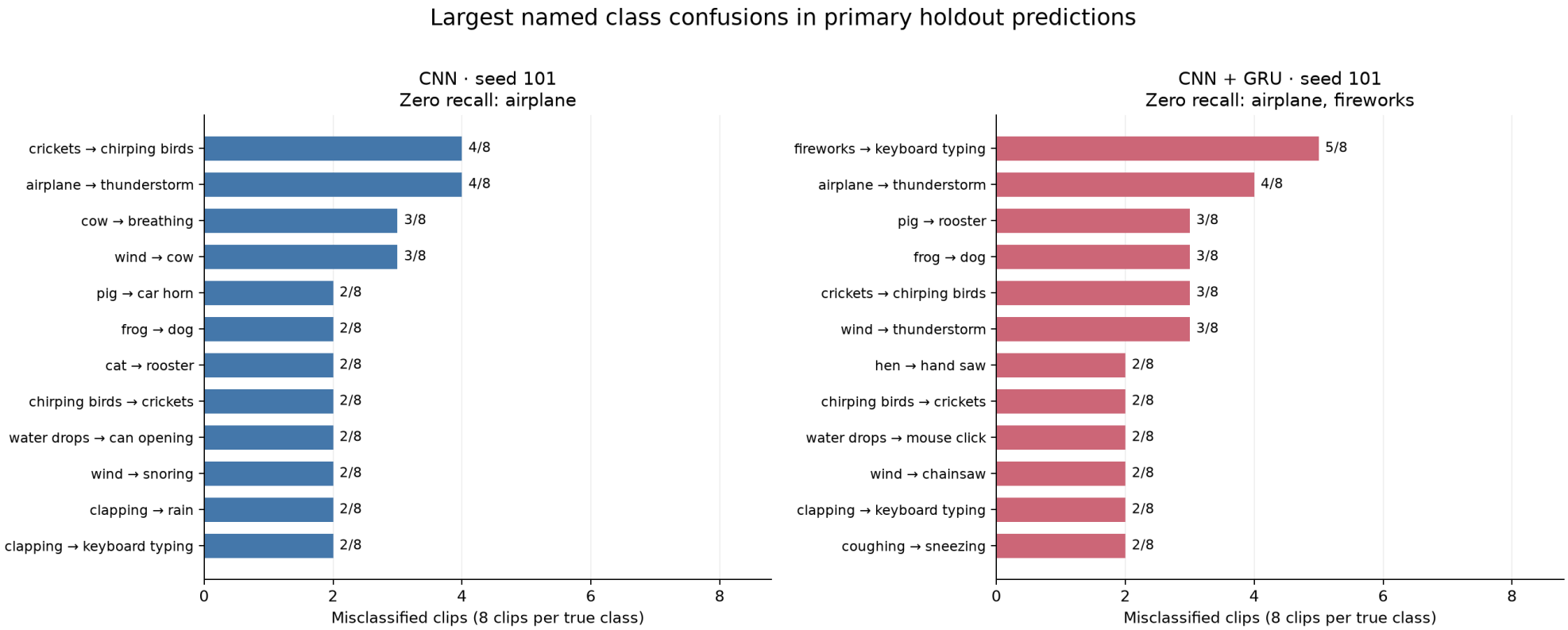


Both curves are measured at epoch end in evaluation mode; online optimization loss is excluded. Train: 1,200 clips; validation: 398 clips.
Checkpoint rule: earliest minimum validation cross-entropy. These are development curves for one seed, not holdout learning curves.

Learning curves

Source: `results/figures/learning_curves.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 3. Named confusions

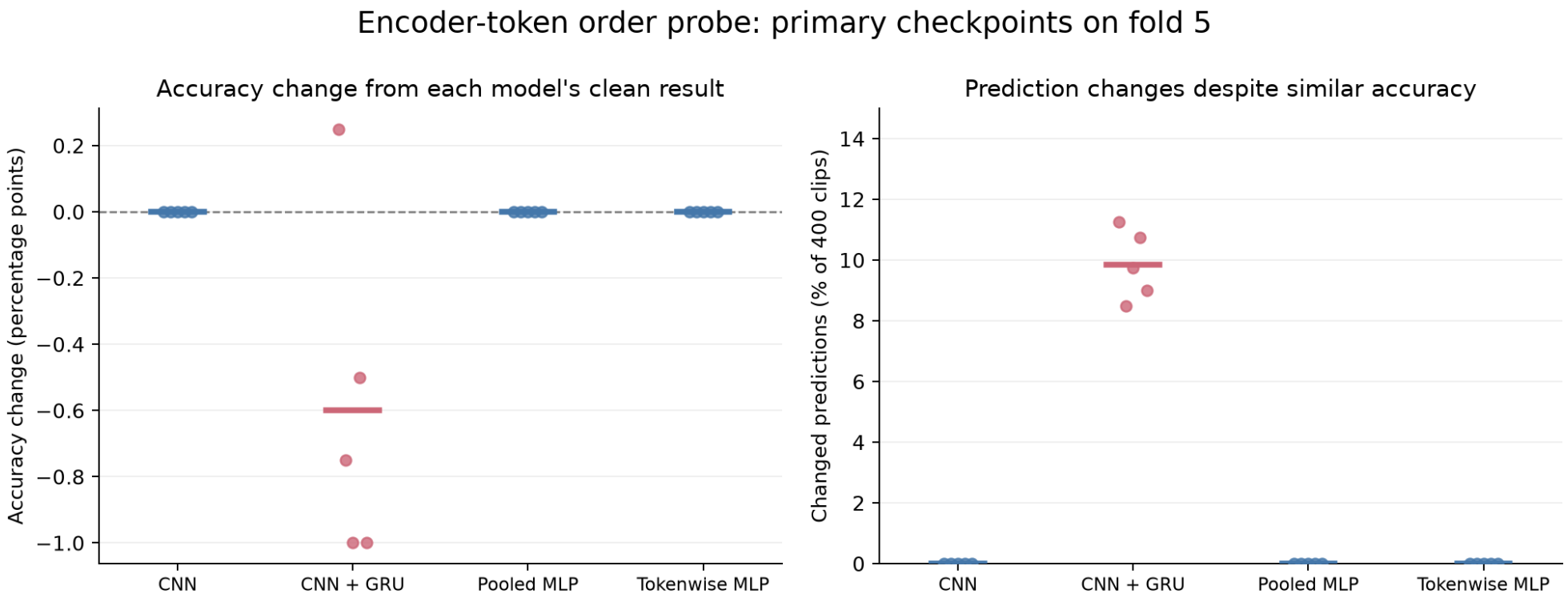


Largest off-diagonal counts, described after evaluation; they were not used to retune the frozen models. A true class may appear in more than one row.
Eight clips per class give coarse estimates. These errors do not characterize every real-world recording condition.

Named confusions

Source: `results/figures/top_confusions.png`. Original saved figure; no experiment was rerun for this PDF.

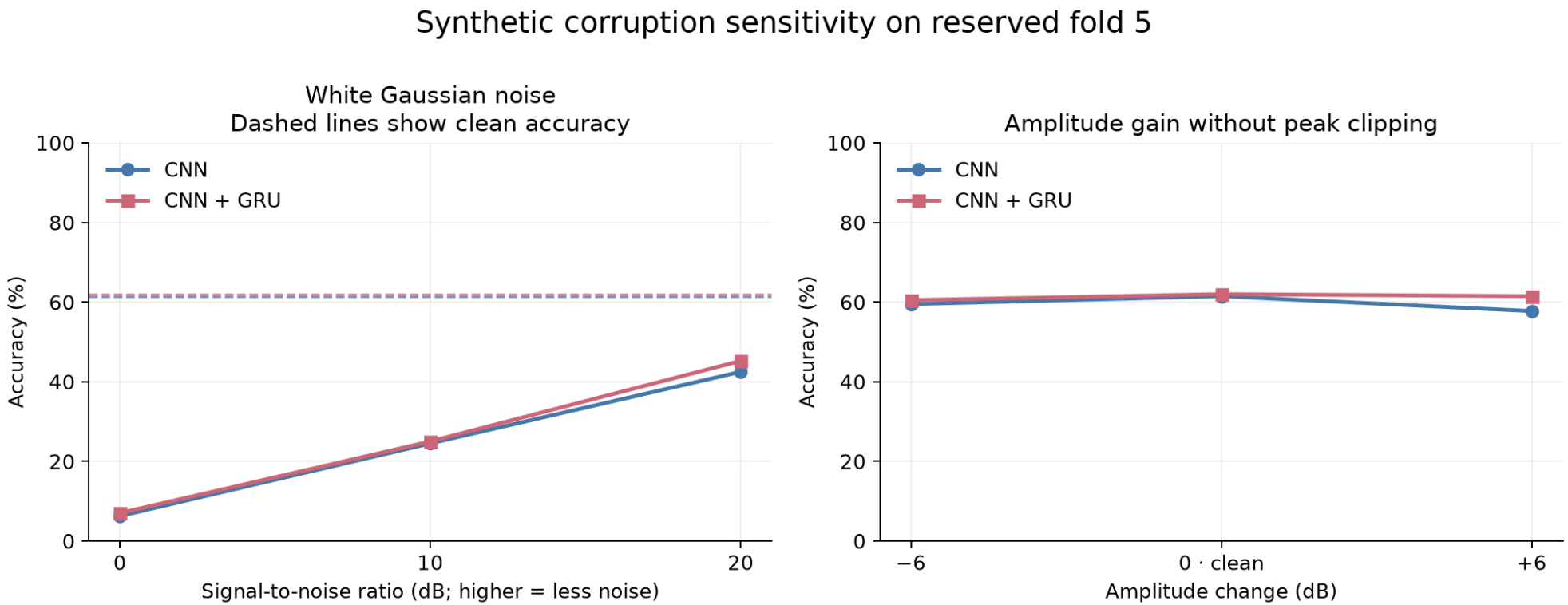
Figure 4. Order probe



Order probe

Source: `results/figures/order_probe.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 5. Synthetic stress

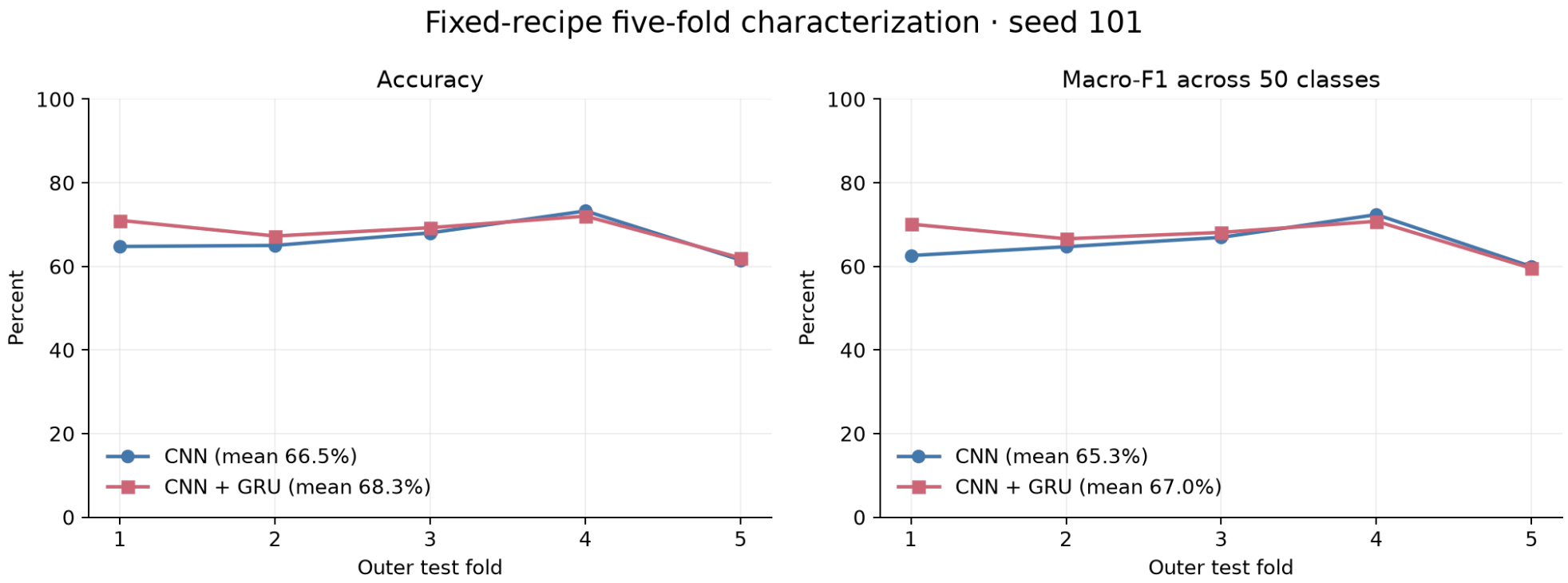


Seed 101 only; same 400 holdout clips. One deterministic noise realization per clip/condition (noise seed 2026); no uncertainty bars.
Synthetic corruption sensitivity does not establish real-world acoustic robustness. Clean reference: CNN 61.5%, CNN + GRU 62.0%.

Synthetic stress

Source: `results/figures/synthetic_stress.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 6. Five-fold characterization

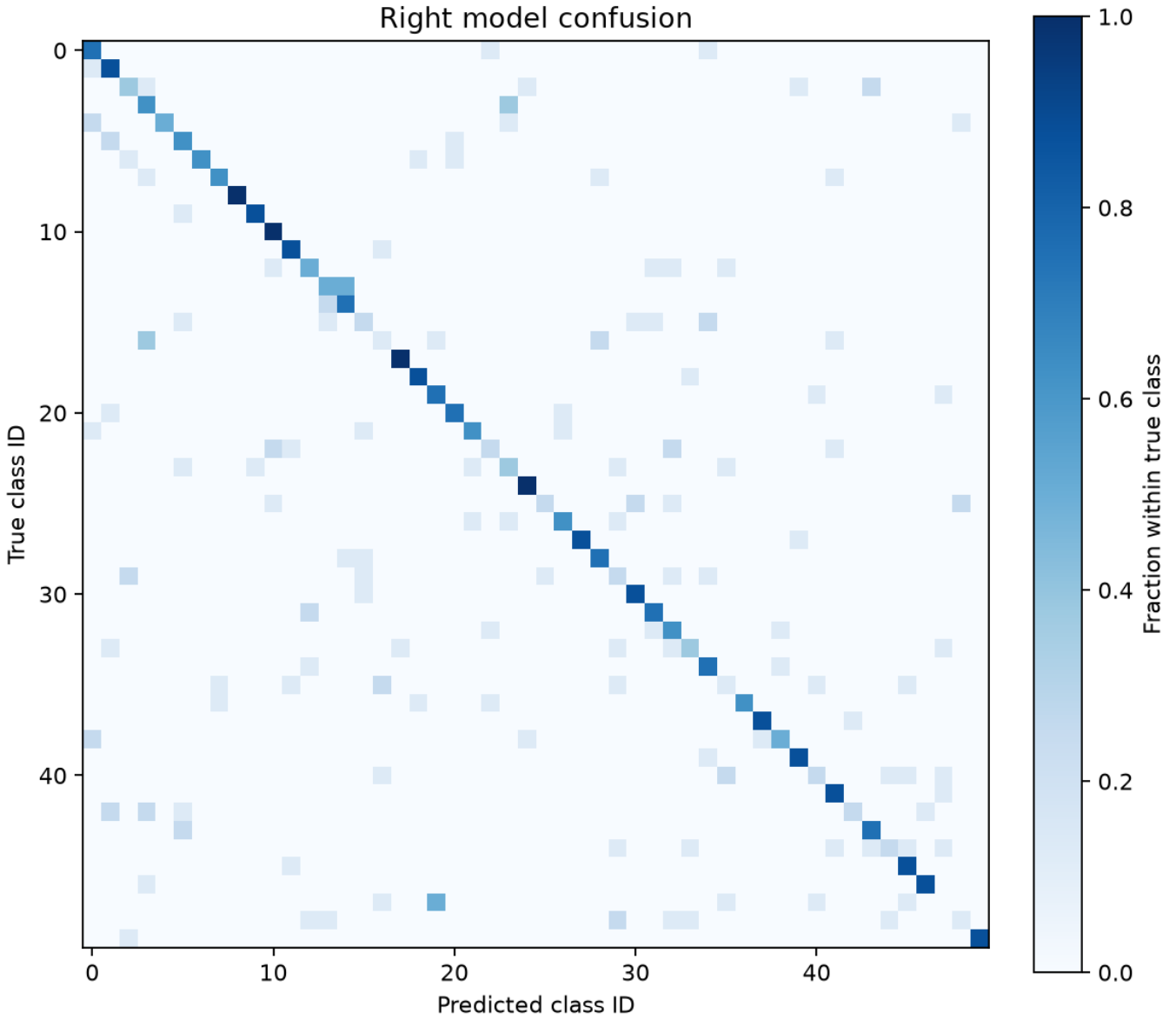


Source-strict rotating folds; one seed. Folds 1–4 already informed development: this is not nested model selection or five untouched confirmations. Fold scores share training data and are dependent; no confidence intervals are shown.

Five-fold characterization

Source: `results/figures/cross_validation.png`. Original saved figure; no experiment was rerun for this PDF.

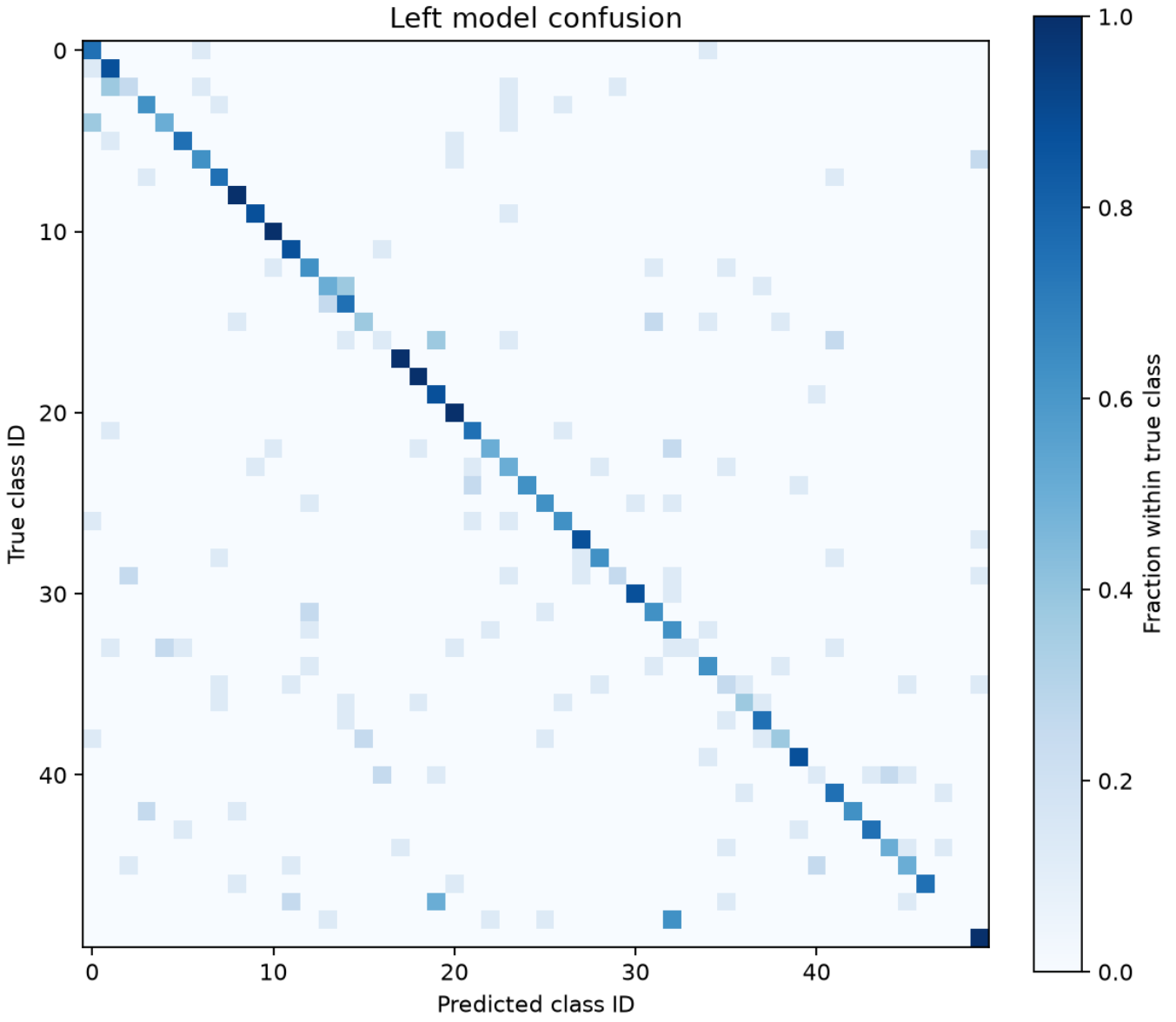
Figure 7. CNN primary fold-5 confusion matrix



Historical primary seed101 checkpoint; 400 clips, eight per class. In the original comparison files, left is CRNN and right is CNN. The plotted class IDs follow the official metadata.

Source: `results/holdout_primary_comparison/right_confusion.png`. Original saved figure; no experiment was rerun for this PDF.

Figure 8. CRNN (CNN + BiGRU) primary fold-5 confusion matrix



Historical primary seed101 checkpoint; 400 clips, eight per class. In the original comparison files, left is CRNN and right is CNN. The plotted class IDs follow the official metadata.

Source: `results/holdout_primary_comparison/left_confusion.png`. Original saved figure; no experiment was rerun for this PDF.